

Simulation of Benzene–Toluene Fractional Distillation using DWSIM

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1. Introduction

Fractional distillation is an important separation process used in chemical and petroleum industries for separating liquid mixtures based on differences in volatility. In this project, a binary mixture of benzene and toluene was separated using a multi-stage distillation column simulated in DWSIM. Benzene acts as the light key component while toluene acts as the heavy key component.

2. Objective

The objective of this project was to design and simulate a 21-stage fractional distillation column for separation of benzene and toluene using the Raoult's Law thermodynamic model in DWSIM and achieve high-purity toluene in the bottom product.

3. Theory of Distillation

Distillation operates on the principle of vapor-liquid equilibrium. The more volatile component vaporizes more readily and rises upward through the column, while the less volatile component remains in the liquid phase and moves downward. Multiple equilibrium stages improve separation efficiency. Reflux plays an important role in increasing product purity.

4. Basic Equations

Overall Material Balance:

$$F = D + B$$

Component Balance for Benzene:

$$Fz_F = Dx_D + Bx_B$$

Reflux Ratio:

$$R = L / D$$

5. Simulation Specifications

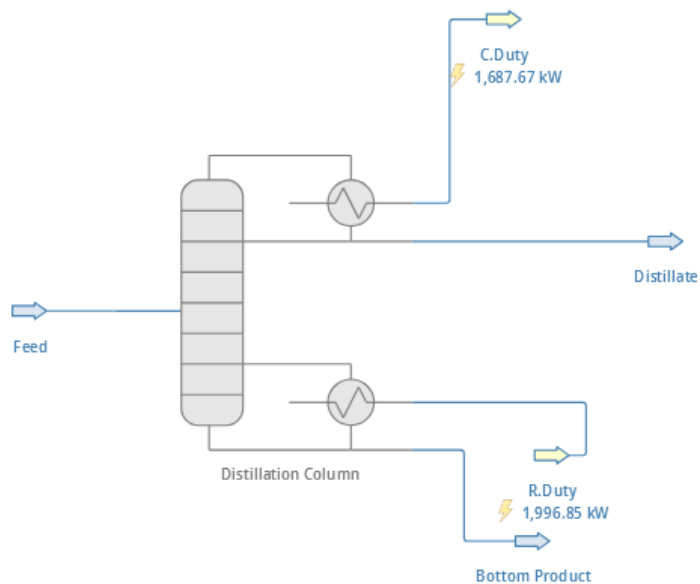
Parameter	Value
Feed Flow Rate	100 kmol/hr
Feed Temperature	25 °C
Feed Pressure	1 atm
Benzene Composition	0.6
Toluene Composition	0.4
Number of Stages	21

Feed Stage	9
Reflux Ratio	2.31
Property Package	Raoult's Law
Condenser Type	Total Condenser

6. Simulation Results

The simulation successfully converged using the Wang-Henke bubble point method in DWSIM. The condenser duty and reboiler duty obtained from the simulation were approximately 1687.67 kW and 1996.85 kW respectively. The bottoms product specification for benzene concentration was achieved successfully.

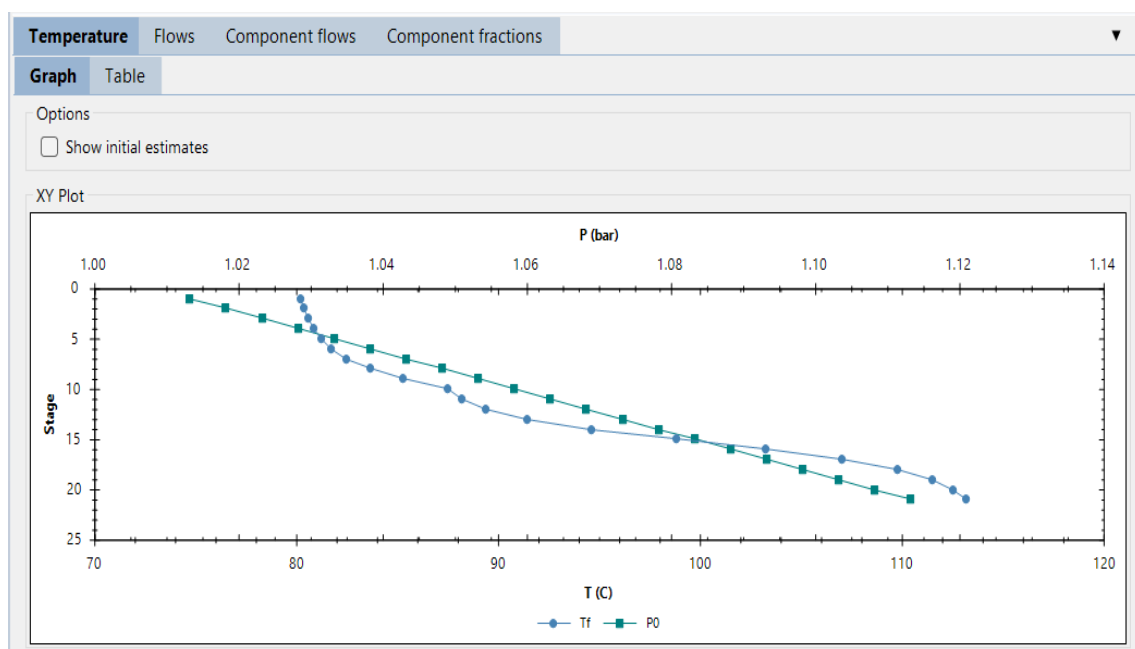
Main Flowsheet



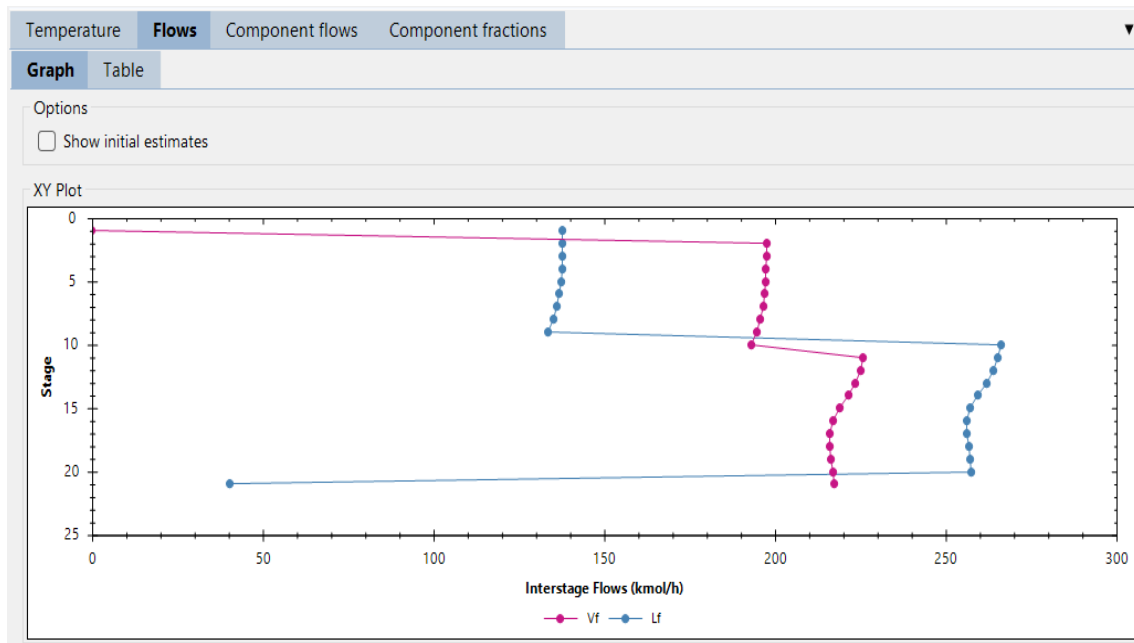
Column Results

General	Specifications	Stages	Connections	Estimates	Results
View Temperature, Pressure and Composition Profiles					
View Properties Profile					
View Convergence Report					
Property		Value	Units		
Condenser Duty		1687.67	kW		
Reboiler Duty		-1996.85	kW		
Condenser Specification Value		2.31			
Condenser Specification Calculated Va...		2.31			
Reboiler Specification Value		0.01			
Reboiler Specification Calculated Value		0.0099999			
Iterations Taken		0			
Estimated Height		11500	mm		

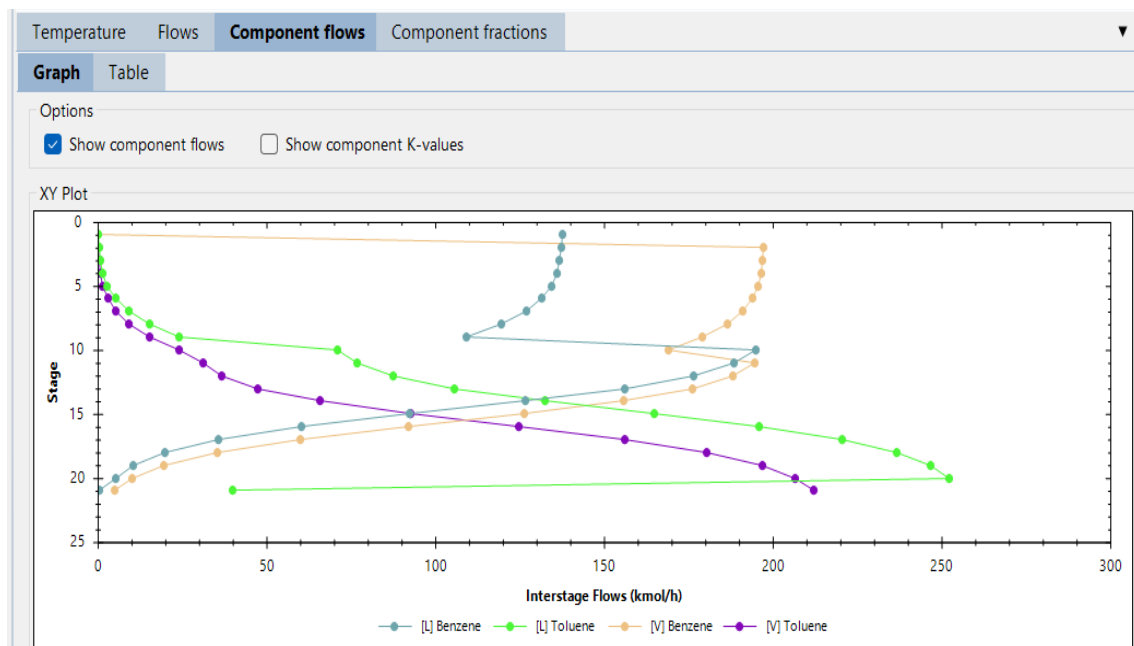
Temperature Profile



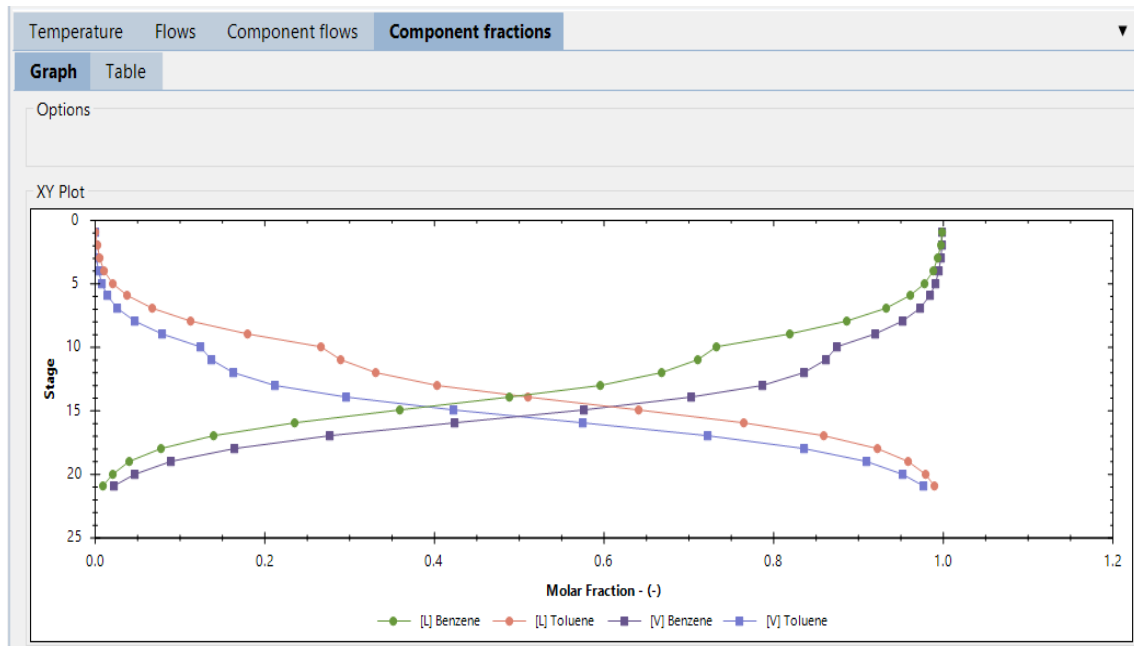
Flow Profile



Component Flow Profile



Component Fraction Profile



7. Conclusion

A rigorous equilibrium-stage simulation of a benzene–toluene fractional distillation column was performed successfully using DWSIM. The simulation demonstrated effective separation of the binary mixture and achieved the desired bottoms purity specification. The project also helped in understanding vapor-liquid equilibrium, reflux operation, stage-wise composition variation, and temperature profiles in a distillation column.

8. Future Scope

Future improvements may include optimization of reflux ratio, energy integration, tray efficiency analysis, pressure variation studies, and comparison with industrial process simulators such as Aspen Plus or HYSYS.

9. References

1. DWSIM Open Source Process Simulator Documentation
2. McCabe, Smith and Harriott – Unit Operations of Chemical Engineering
3. Treybal – Mass Transfer Operations
4. Vapor-Liquid Equilibrium and Distillation Theory Notes